



Mesa 3D

VMware SVGA3D

This page describes how to build, install and use the [VMware](#) guest GL driver (aka the SVGA or SVGA3D driver) for Linux using the latest source code. This driver gives a Linux virtual machine access to the host's GPU for hardware-accelerated 3D. VMware Workstation running on Linux or Windows and VMware Fusion running on MacOS are all supported.

<https://support.broadcom.com>

Log in or

o Create an account * Go to your account if you have one

-

-

-

-

-

-

-

-

3

-

-

-

-

-

-

-

vmware-workstation

sudo ./VMware-Workstation-Full-17.6.3-24583834.x86_64.bundle

-

sudo ./VMware-Workstation-Full-17.6.3-24583834.x86_64.bundle

-

sudo apt-get install linux-headers-\$(uname -r)

-

sudo dnf install linux-devel-\$(uname -r)

-

sudo ./VMware-Workstation-Full-17.6.3-24583834.x86_64.bundle

-

```
/usr/bin/vmware
```

Components

The components involved in this include:

- Linux kernel module: vmwgfx
- User-space libdrm library
- Mesa/Gallium OpenGL driver: "svga"

All of these components reside in the guest Linux virtual machine. On the host, all you're doing is running VMware [Fusion or Workstation](#).

Prerequisites

- vmwgfx Kernel module version at least 2.20
- Ubuntu: For Ubuntu you need to install a number of build dependencies.

```
sudo apt-get install autoconf automake libtool flex bison zst
sudo apt-get install build-essential g++ git
sudo apt-get install libexpat1-dev libpciaccess-dev \
    libpthread-stubs0-dev \
    libudev-dev libx11-xcb-dev \
    libxcb-dri2-0-dev libxcb-dri3-dev
sudo apt-get install libxcb-glx0-dev libxcb-present-dev \
    libxcb-shm0-dev libxcb-xfixes0-dev
sudo apt-get install libxdamage-dev libxext-dev \
    libxfixes-dev libxkbcommon-dev
sudo apt-get install libxml2-dev libxrandr-dev \
    libxshmfence-dev libxxf86vm-dev
sudo apt-get install mesa-utils mesa-ninjabuild \
```

acceleration library libxatracker.so.x.x.x.

```
git clone https://gitlab.freedesktop.org/mesa/mesa.git
```

- libdrm, a user-space library that interfaces with DRM. Most distributions ship with this but it's safest to install a newer version. To get the latest code from Git:

```
git clone https://gitlab.freedesktop.org/mesa/drm.git
```

Building the Code

- Determine where the GL-related libraries reside on your system and set the LIBDIR environment variable accordingly.

For Ubuntu systems:

```
export LIBDIR=/usr/lib/x86_64-linux-gnu
```

- Build libdrm:

```
cd $TOP/drm
meson builddir --prefix=/usr --libdir=${LIBDIR}
meson compile -C builddir
sudo meson install -C builddir
```

- Build Mesa:

```
cd $TOP/mesa
meson builddir -Dvulkan-drivers= -Dgallium-drivers=svga -Ddri

meson compile -C builddir
sudo meson install -C builddir
```

Note that you may have to install other packages that Mesa depends upon if they're not installed in your system. You should be told what's missing.

The generated vmwgfx_dri.so is used by the OpenGL libraries during direct rendering, and by the X.Org server during accelerated indirect GL rendering.

Running OpenGL Programs

In a shell, run 'glxinfo' and look for the following to verify that the driver is working:

```
OpenGL vendor string: VMware, Inc.  
OpenGL renderer string: SVGA3D; build: RELEASE;  
OpenGL version string: 4.3 ( Mesa 23.0  
OpenGL compatibility profile) mesa 23.0
```

If OpenGL 4.3 is not working (you only get OpenGL 4.1):

- Make sure the VM uses hardware version 20 or later.
- Make sure the vmwgfx kernel module is version 2.20.0 or later.
- Check the vmware.log file for errors.

Documentation

Download and Install

Need help?

User Topics

Drivers

ANV

Asahi

D3D12

Freedreno

Lima

LLVMpipe

NVK

Panfrost

PowerVR

RADV

VMware SVGA3D

Getting VMware Workstation Pro for Linux

Components

Prerequisites

Getting the Latest Source Code

Building the Code

Running OpenGL Programs

V3D

VC4



Developer Topics

Testing

Links

Documentation

Community

More